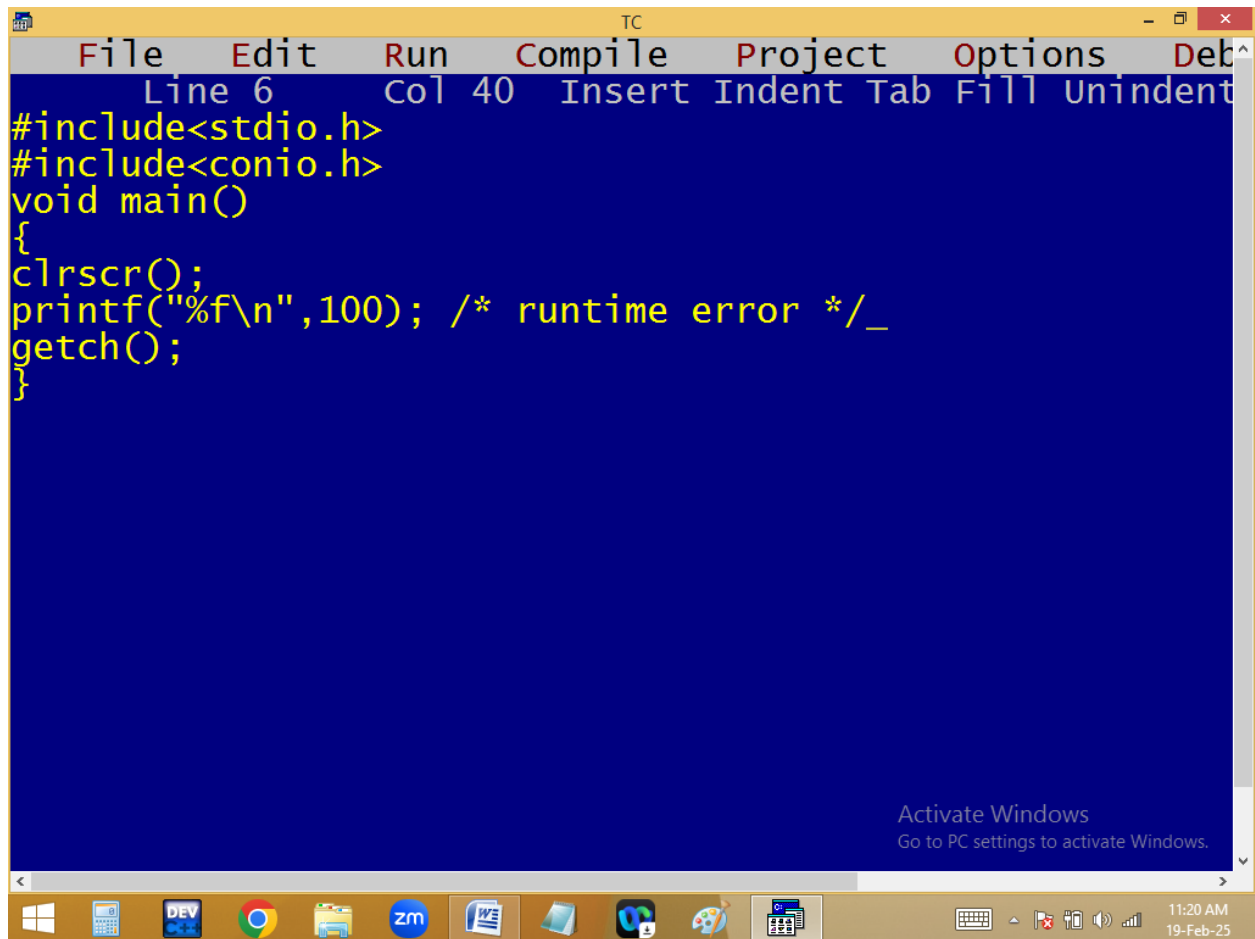
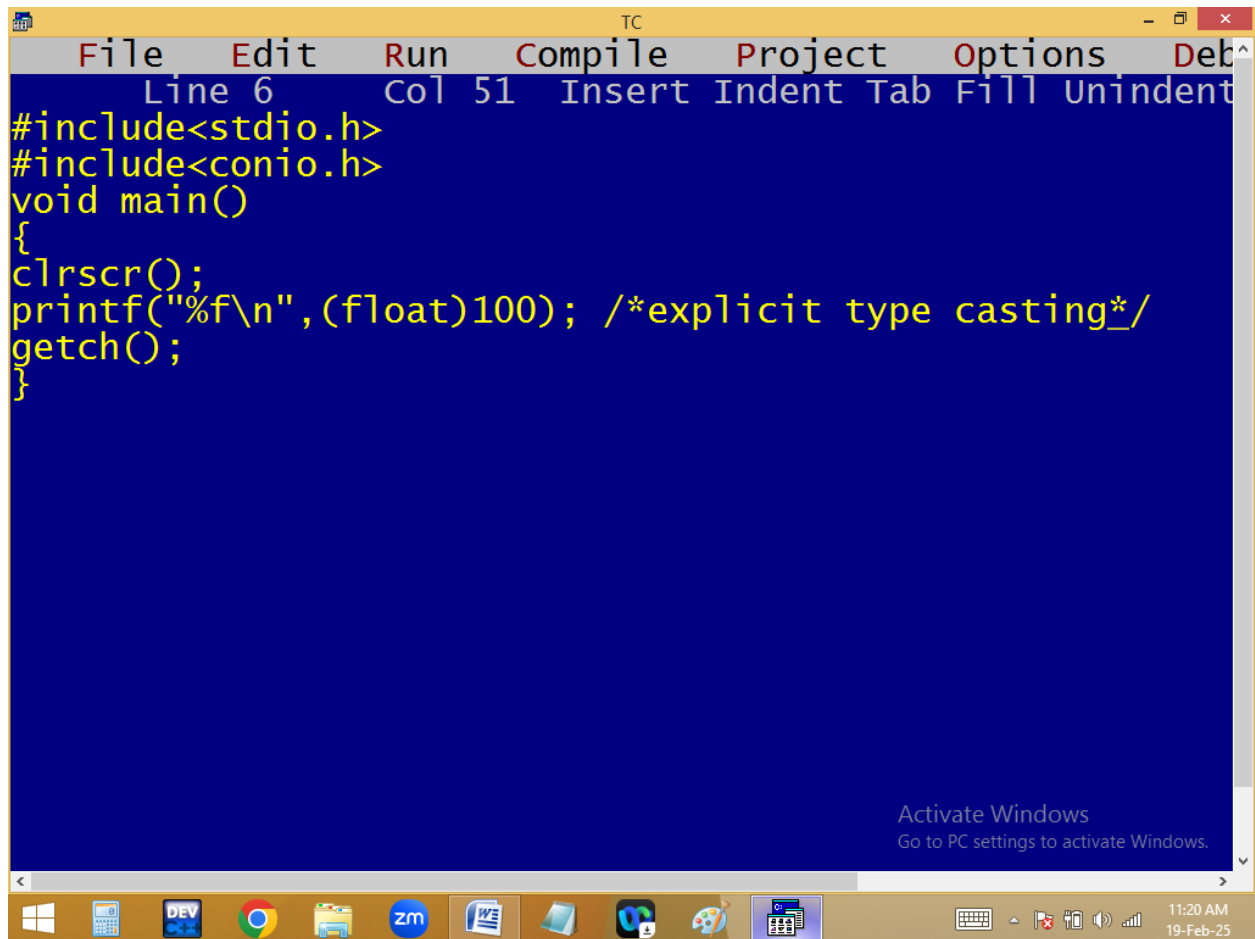




The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 40", and lists editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",100); /* runtime error */_
getch();
}
```

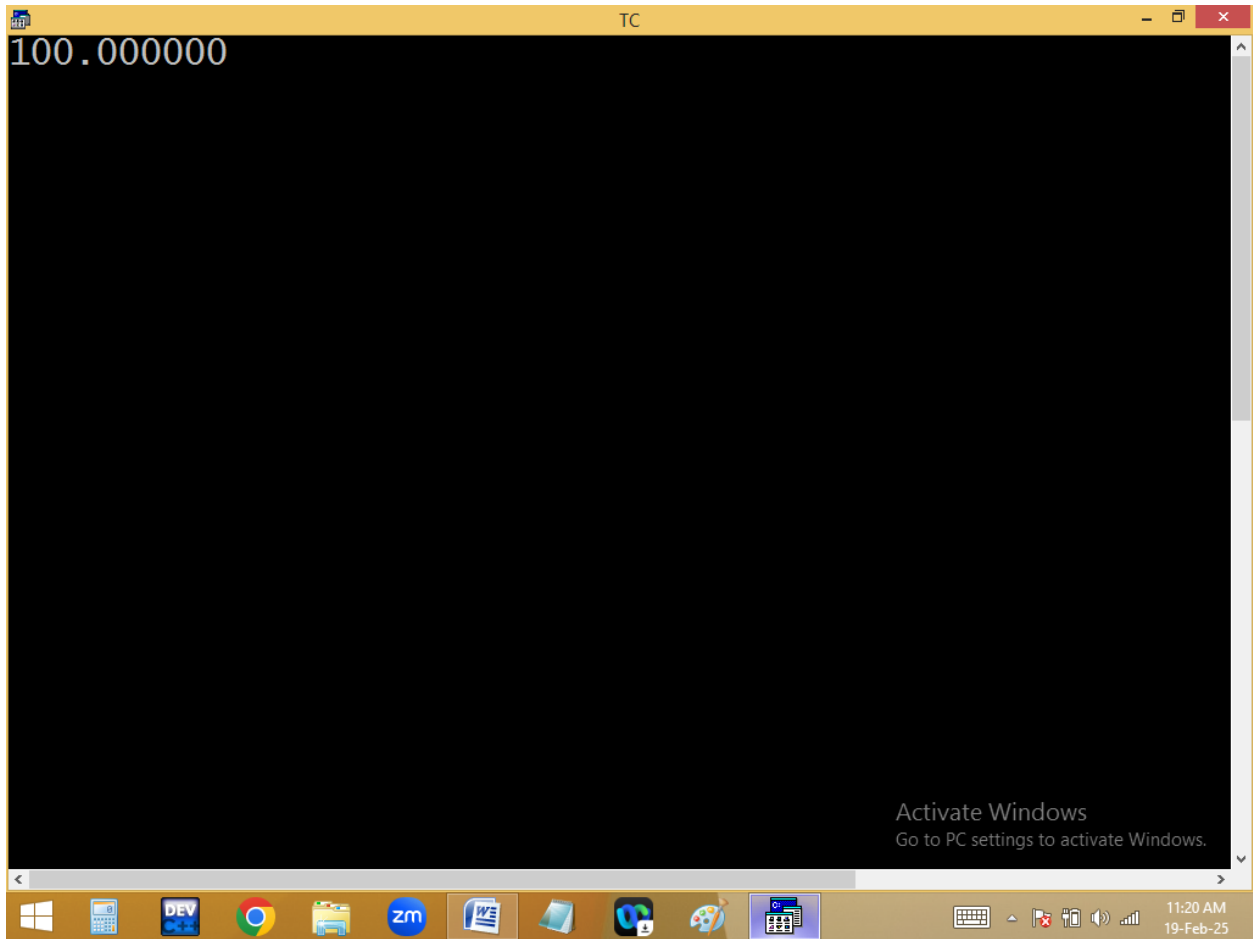
In the bottom right corner of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and other applications. The system tray on the right shows the time as 11:20 AM and the date as 19-Feb-25.

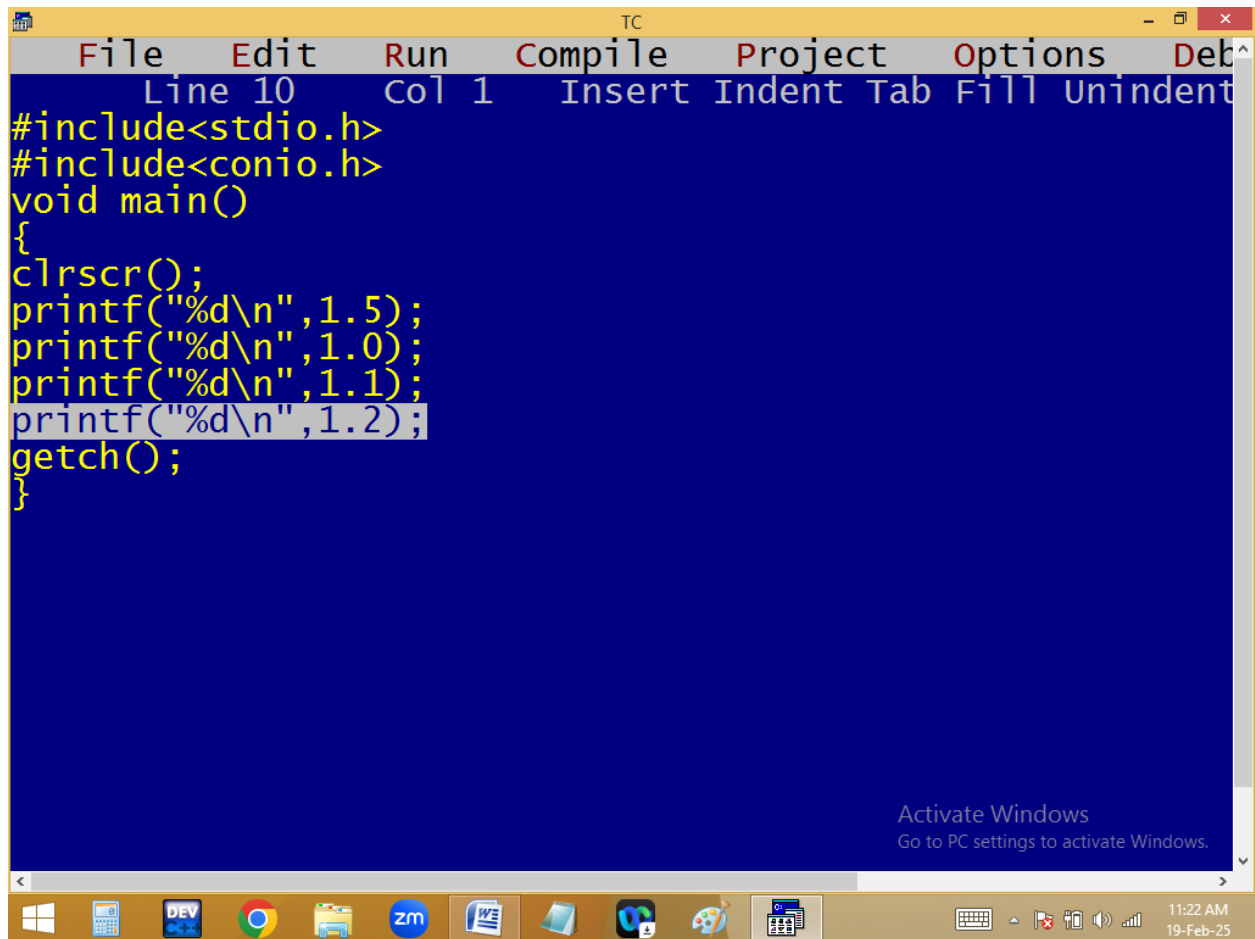


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 51", and lists editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",(float)100); /*explicit type casting*/
getch();
}
```

An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, stating "Go to PC settings to activate Windows." The Windows taskbar is at the bottom, showing icons for the Start menu, Task View, Calculator, Visual Studio Code, Google Chrome, File Explorer, Zoom, Word, Edge, and Paint. The system tray on the right shows the date and time as "11:20 AM 19-Feb-25".

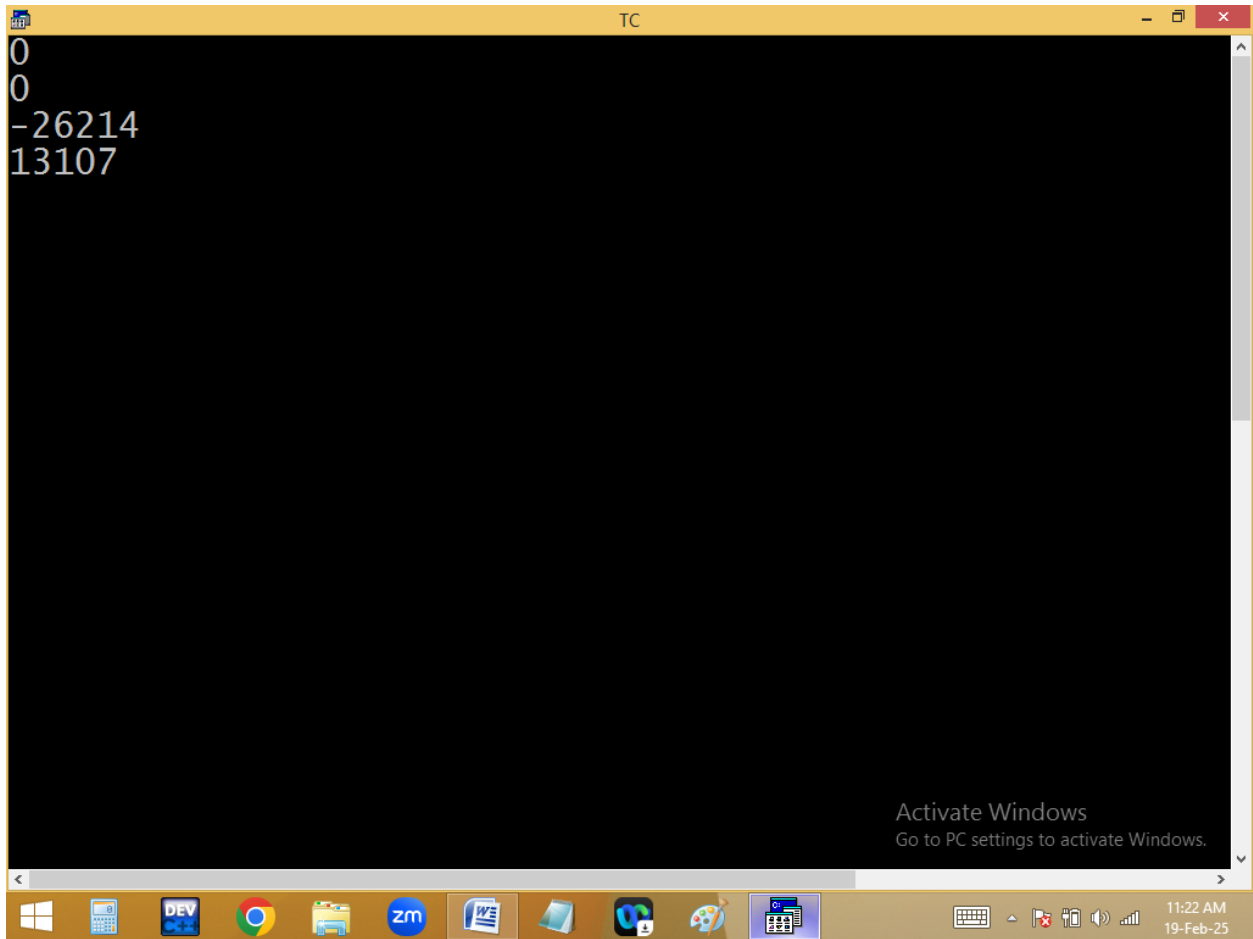


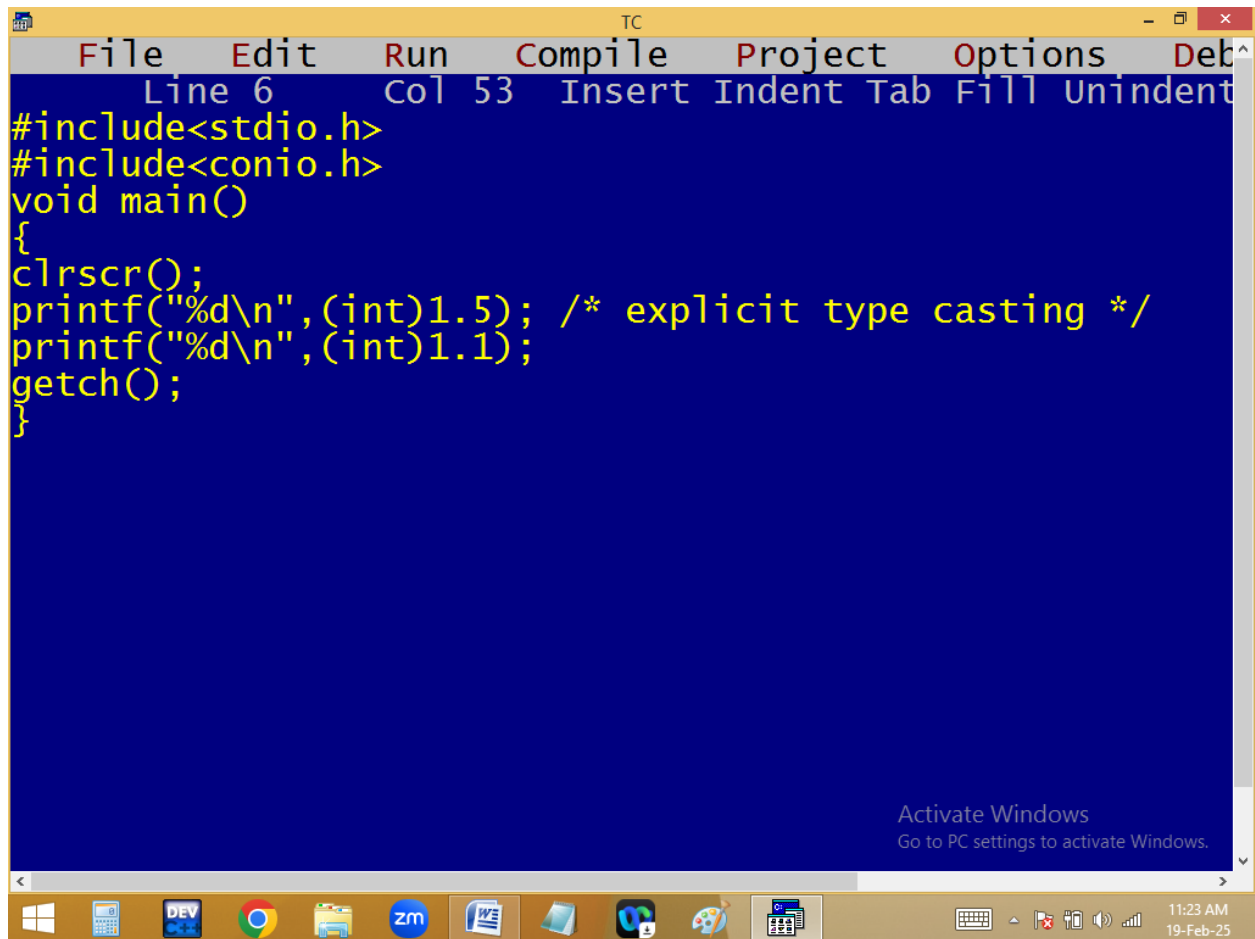


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 10", "Col 1", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",1.5);
printf("%d\n",1.0);
printf("%d\n",1.1);
printf("%d\n",1.2);
getch();
}
```

The line containing `printf("%d\n",1.2);` is highlighted. In the bottom right corner of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock in the bottom right corner shows "11:22 AM" and "19-Feb-25".

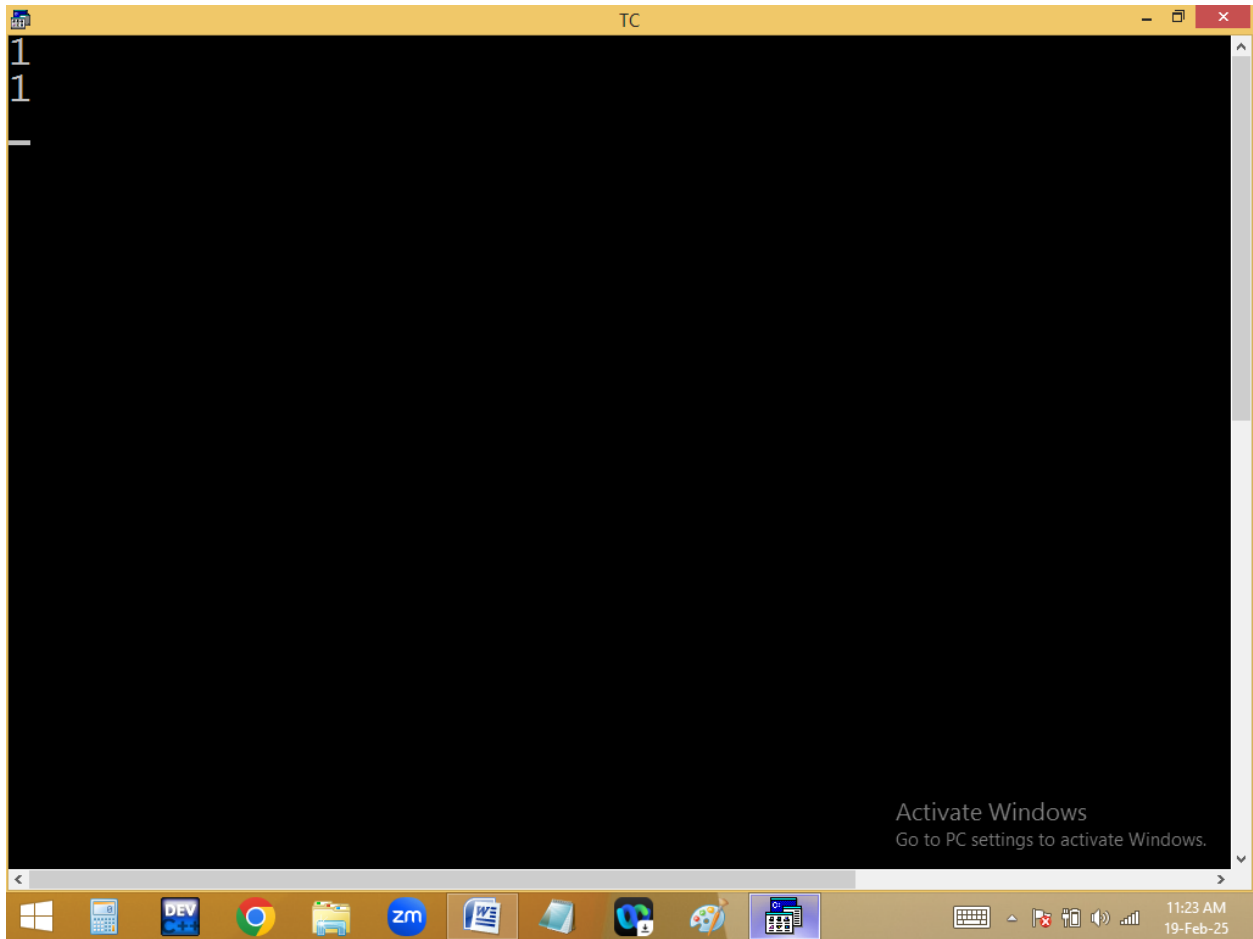


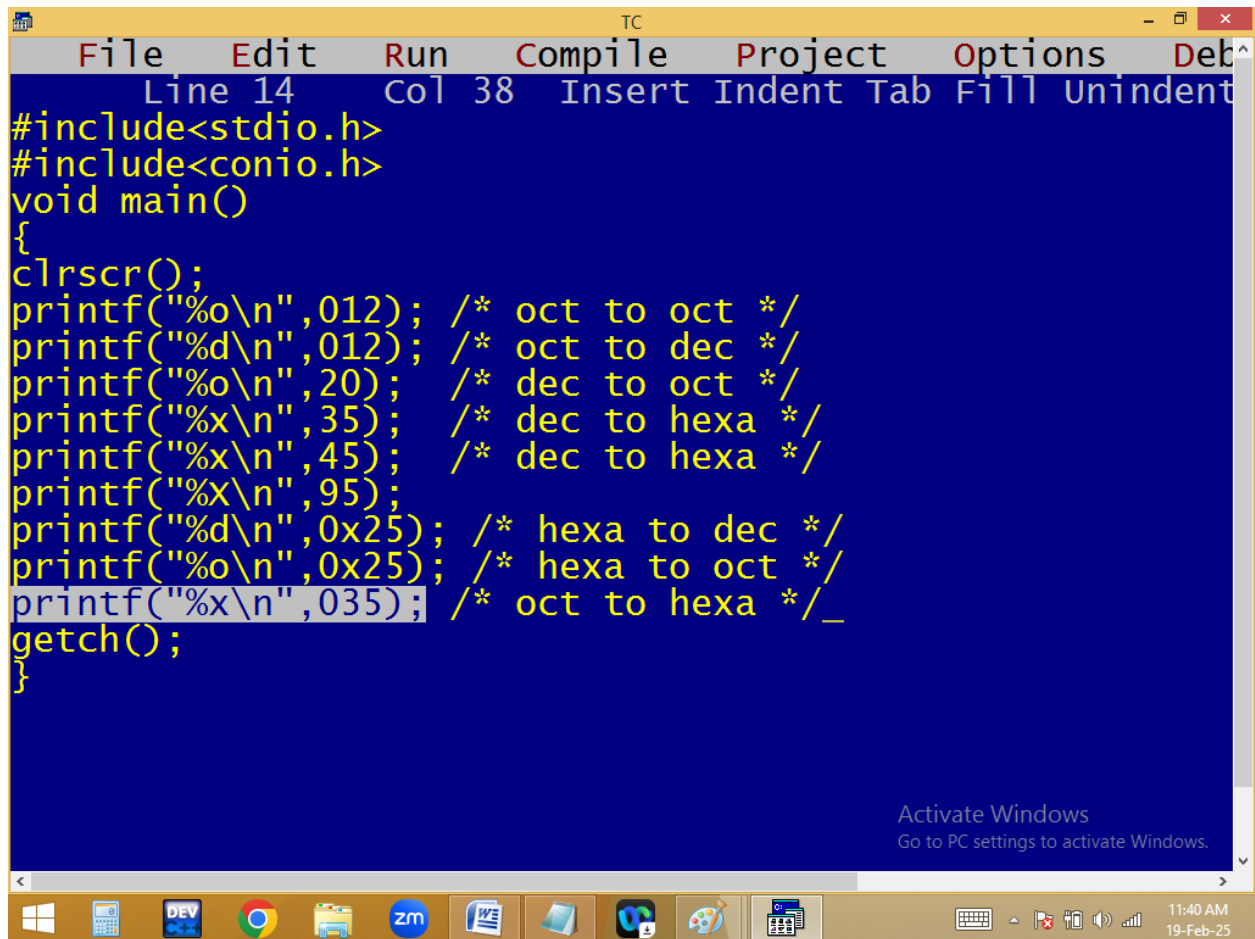


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below it is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 6", "Col 53", and a list of editing actions: Insert, Indent, Tab, Fill, and Unindent. The main editor area has a dark blue background and contains the following C code in yellow text:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",(int)1.5); /* explicit type casting */
printf("%d\n",(int)1.1);
getch();
}
```

In the bottom right corner of the editor area, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom of the screen, showing various application icons including Windows Explorer, Dev C++, Google Chrome, File Explorer, Zoom, Word, and others. The system clock in the bottom right corner of the taskbar shows "11:23 AM" and "19-Feb-25".

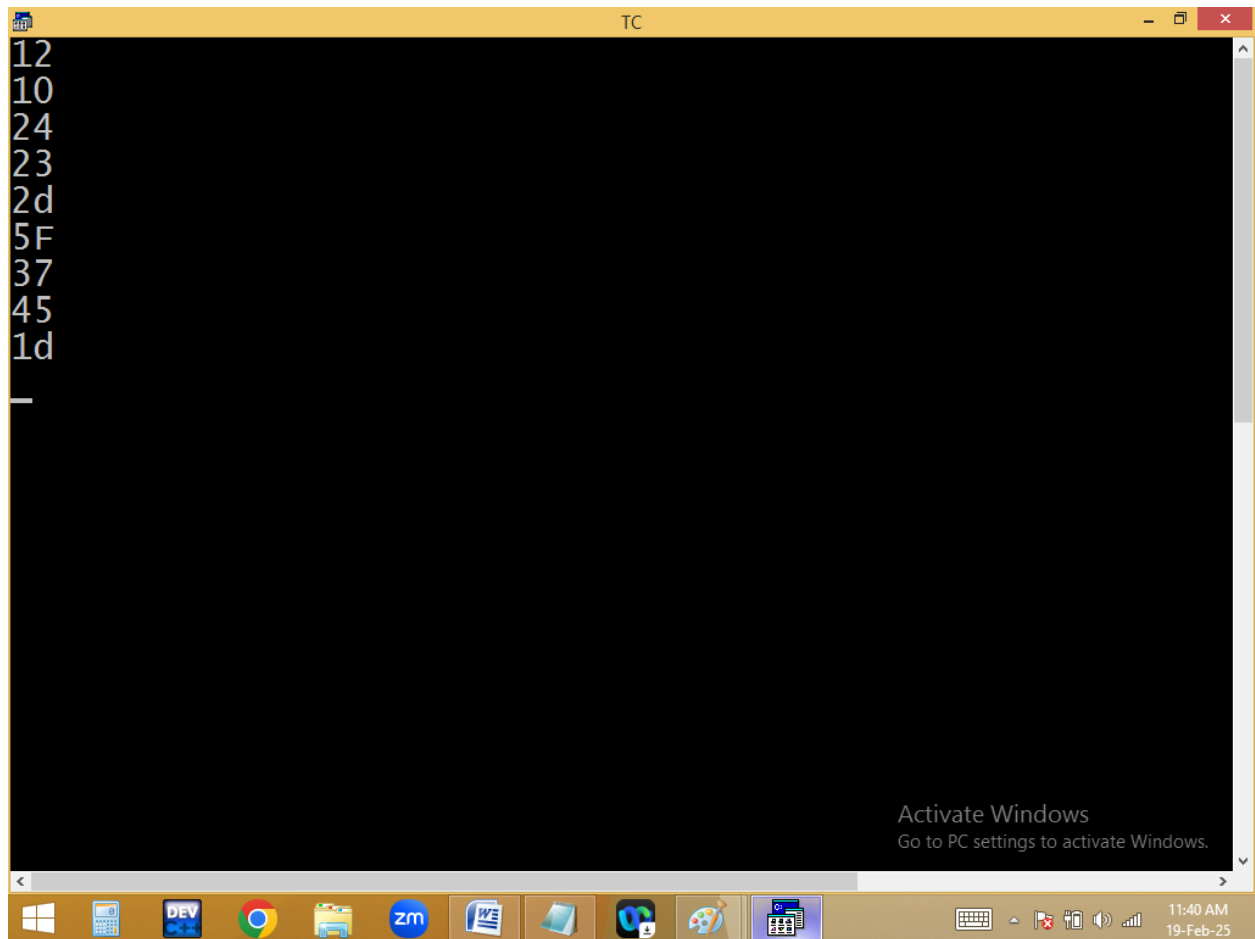




```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 38 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%o\n",012); /* oct to oct */
printf("%d\n",012); /* oct to dec */
printf("%o\n",20); /* dec to oct */
printf("%x\n",35); /* dec to hexa */
printf("%x\n",45); /* dec to hexa */
printf("%X\n",95);
printf("%d\n",0x25); /* hexa to dec */
printf("%o\n",0x25); /* hexa to oct */
printf("%x\n",035); /* oct to hexa */
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

11:40 AM
19-Feb-25



octal to decimal

$$\begin{array}{r} 012 \\ \swarrow \quad \searrow \\ 8 \times 1 + 8 \times 2 \\ \hline 8 + 16 = 24 \end{array}$$

dec to octal

$$\begin{array}{r} 8 \overline{) 20} \\ 2-4 \checkmark \end{array}$$

dec to hexa

$$\begin{array}{r} 16 \overline{) 35} \\ 2-3 \checkmark \end{array}$$

hexadecimal - 0 to 15 ==> 0-9, 10-a, 11-b, 12-c, 13-d, 14-e, 15-f

dec to hexa

$$\begin{array}{r} 16 \overline{) 45} \\ 2-13 \end{array}$$

dec to hexa

$$\begin{array}{r} 16 \overline{) 95} \\ 5-15 \end{array}$$

hexa to dec

0x25

$$\begin{array}{r} 16^1 \times 2 + 16^0 \times 5 \\ \hline 32 + 5 = 37 \end{array}$$

hex to dec

0x25

$$\begin{array}{r} 16^1 \times 2 + 16^0 \times 5 \\ \hline 32 + 5 = 37 \end{array}$$

dec to oct

$$\begin{array}{r} 8 \overline{) 37} \\ 4-5 \checkmark \end{array}$$

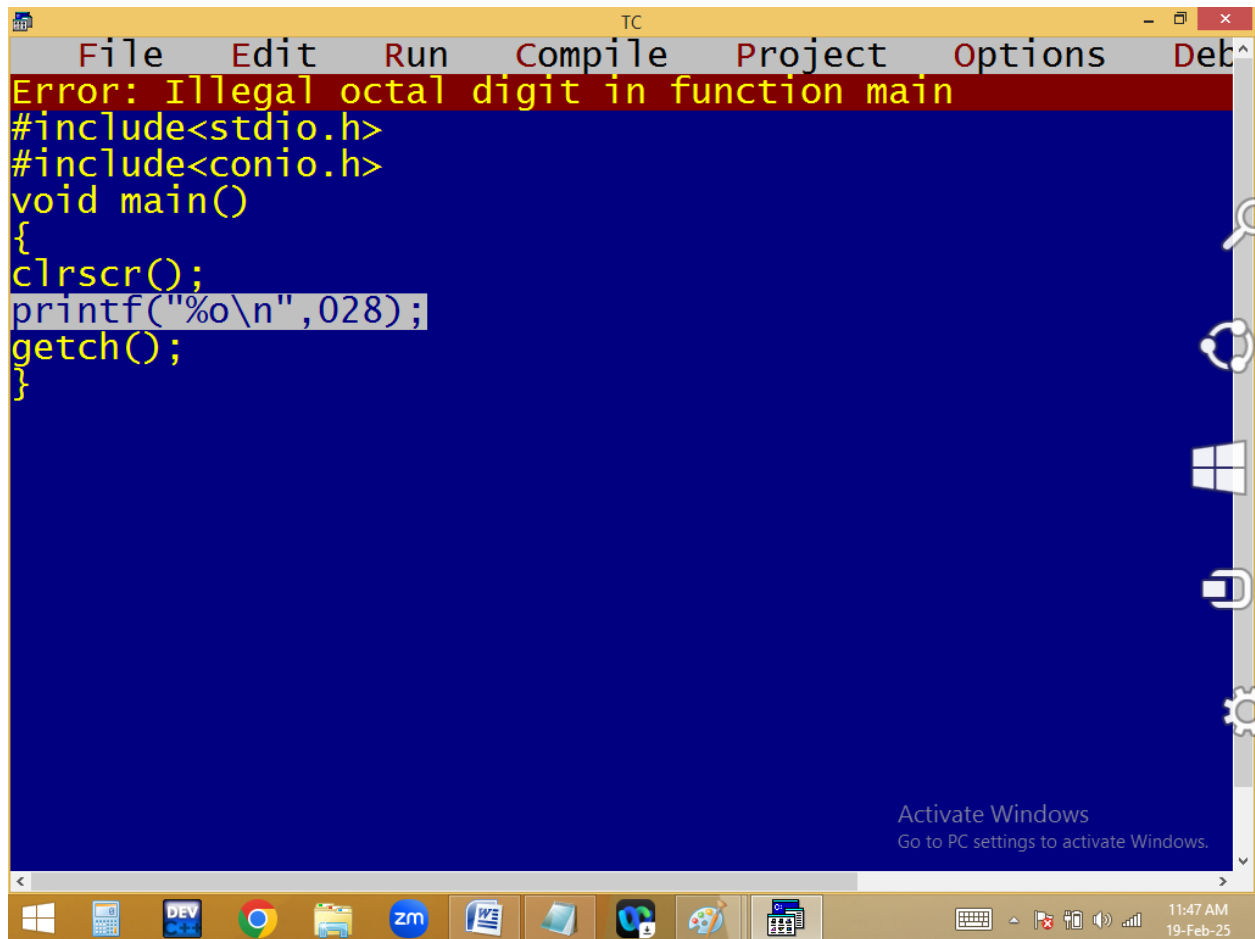
oct to dec

035

$$\begin{array}{r} 8^1 \times 3 + 8^0 \times 5 \\ \hline 24 + 5 = 29 \end{array}$$

dec to hexa

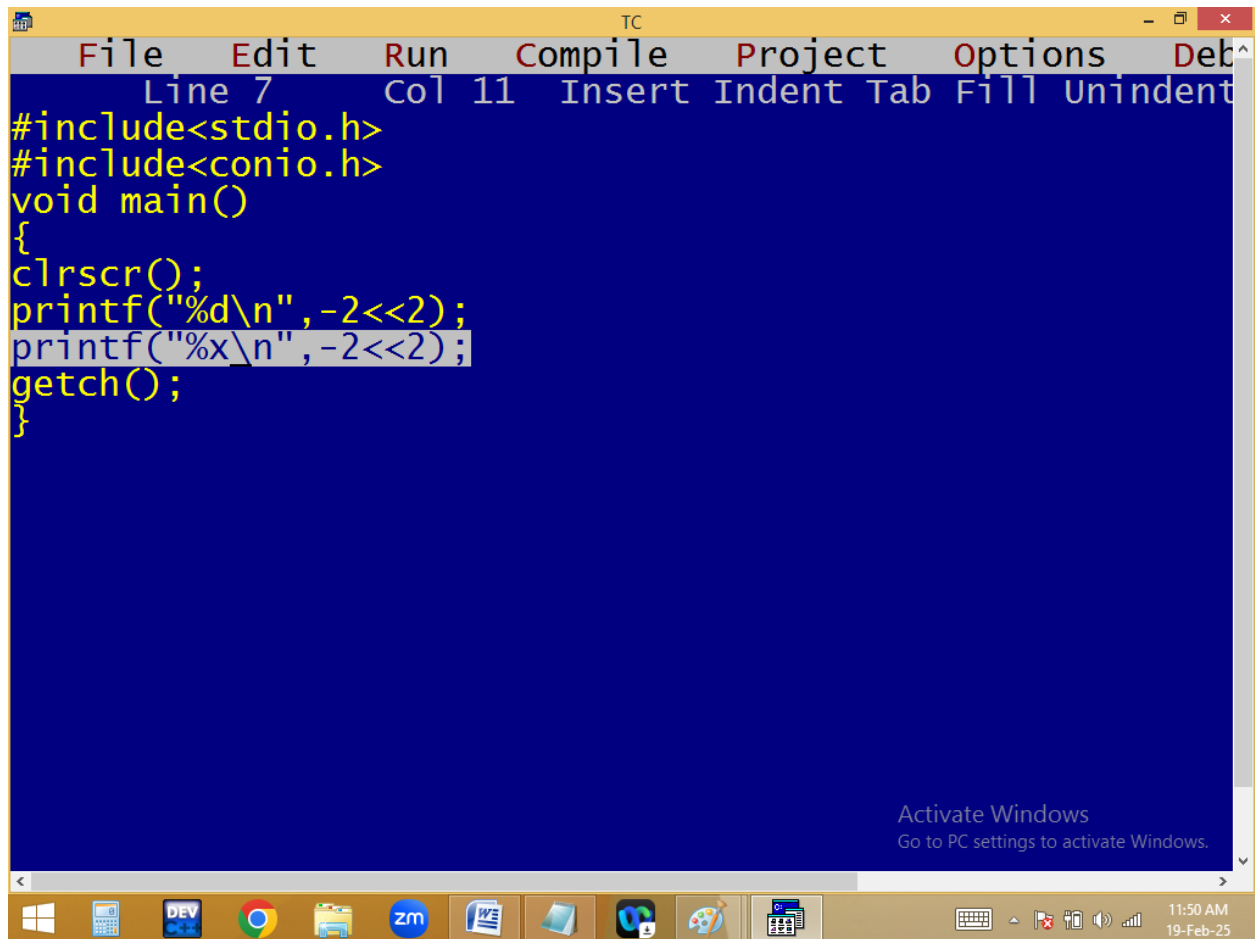
$$\begin{array}{r} 16 \overline{) 29} \\ 1-13 \\ \hline d \checkmark \end{array}$$



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top states: "Error: Illegal octal digit in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%o\n",028);
getch();
}
```

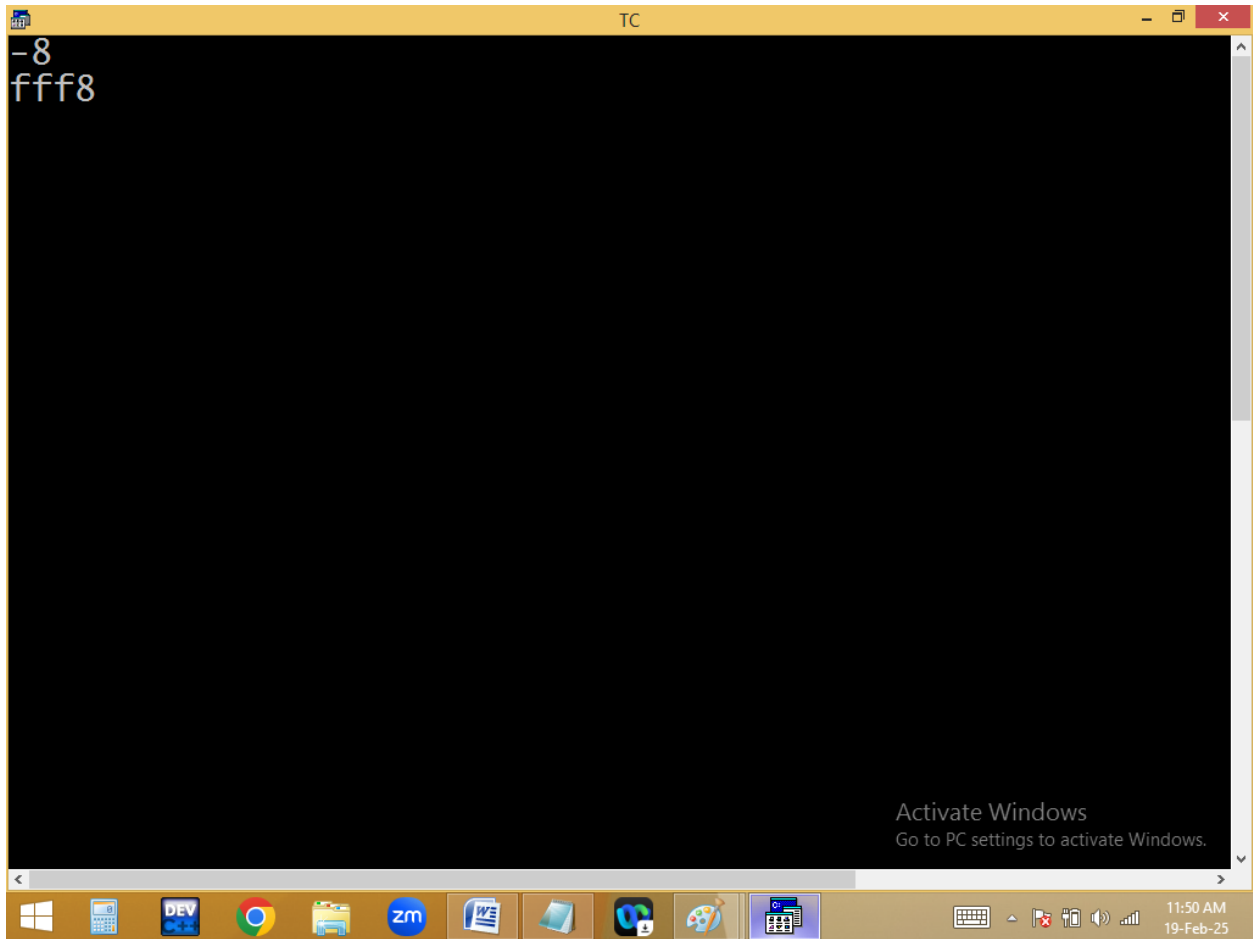
The line `printf("%o\n",028);` is highlighted. The Windows taskbar is visible at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system tray on the right shows the date and time as "11:47 AM 19-Feb-25". A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.



```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 11 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",-2<<2);
printf("%x\n",-2<<2);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

11:50 AM
19-Feb-25



$$2 \ll 2 = 8$$

$$8 = 0000\ 0000\ 0000\ 1000$$

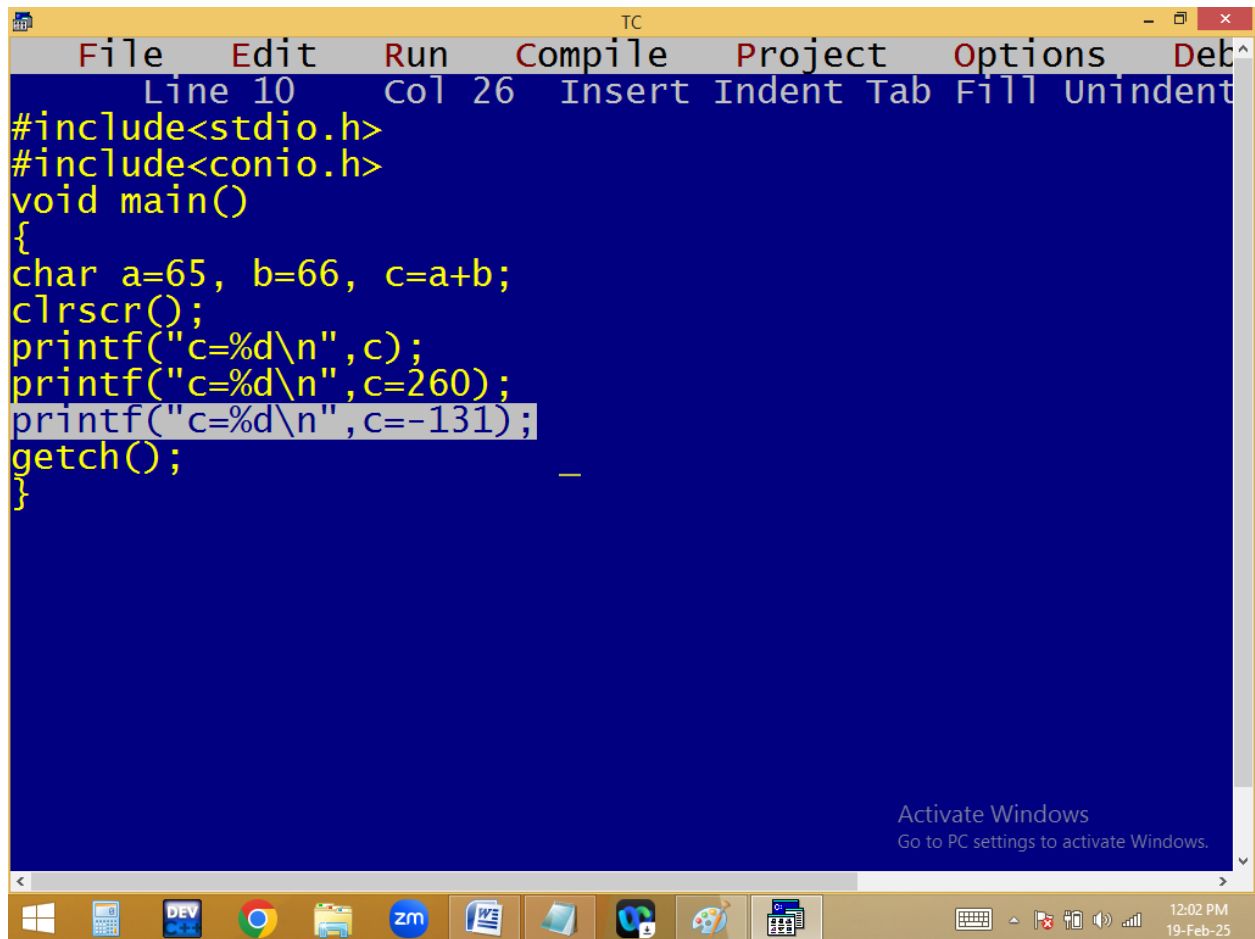
$$1 \sim 1111\ 1111\ 1111\ 0111$$

$$2 \sim 0000\ 0000\ 0000\ 0001$$

$$\begin{array}{r}
 \begin{array}{cccc}
 & & & 111 \\
 \hline
 1111 & 1111 & 1111 & | 000 \\
 \hline
 15 & 15 & 111 & \backslash 3 \\
 8 & 4 & 2 & 1 \quad 8 & 4 & 2 & 1 \quad 8 & 4 & 2 & 1 \quad 2 \\
 \hline
 15 & 15 & 15 & & & & & 8 \\
 \hline
 \uparrow & \uparrow & \uparrow & & & & &
 \end{array}
 \end{array}$$

dec to bin

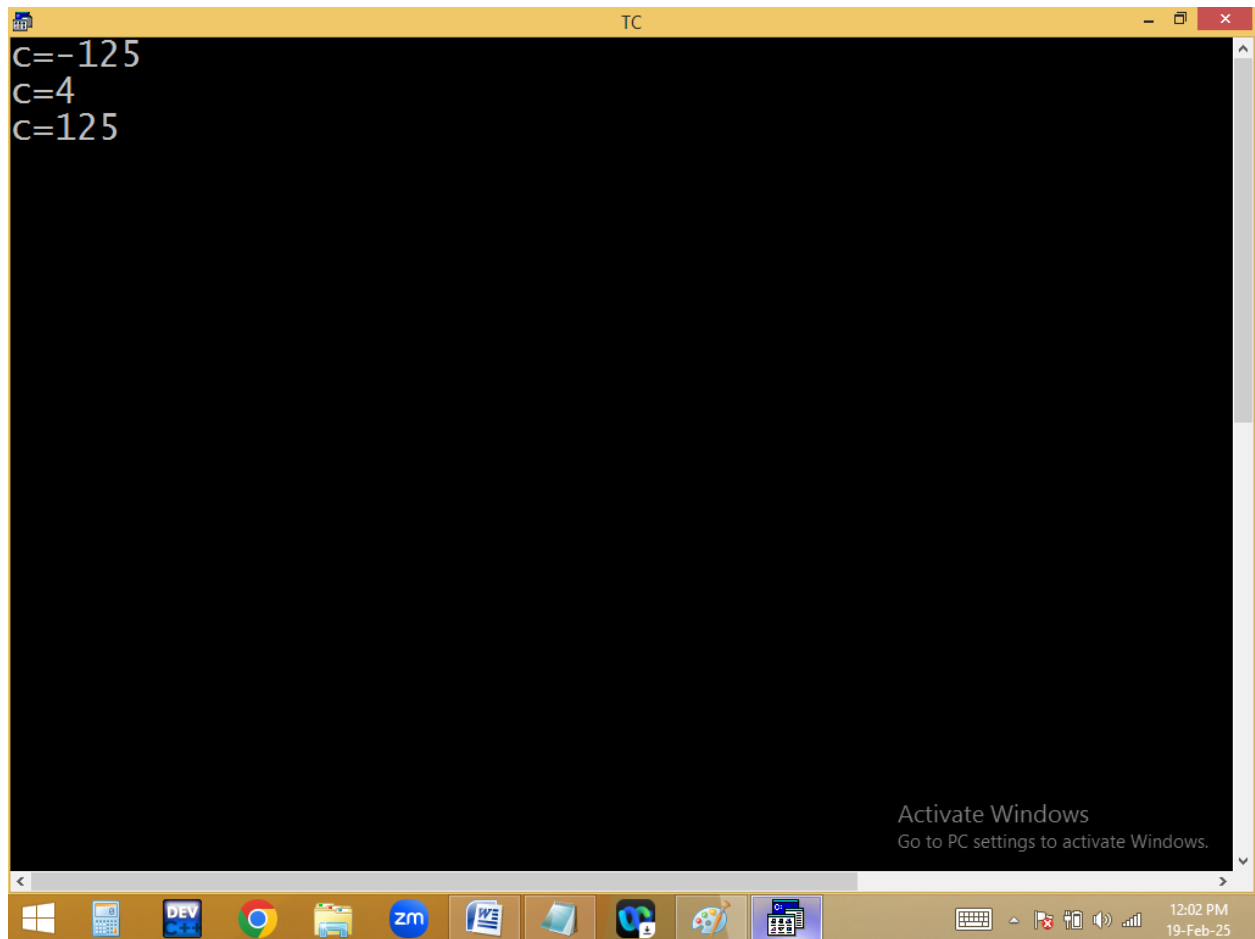
$$\begin{array}{r}
 2 \mid 8 \\
 2 \mid 4 - 0 \\
 2 \mid 2 - 0 \\
 \boxed{1 - 0}
 \end{array}$$



```
TC
File Edit Run Compile Project Options Deb
Line 10 Col 26 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
char a=65, b=66, c=a+b;
clrscr();
printf("c=%d\\n",c);
printf("c=%d\\n",c=260);
printf("c=%d\\n",c=-131);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

12:02 PM
19-Feb-25

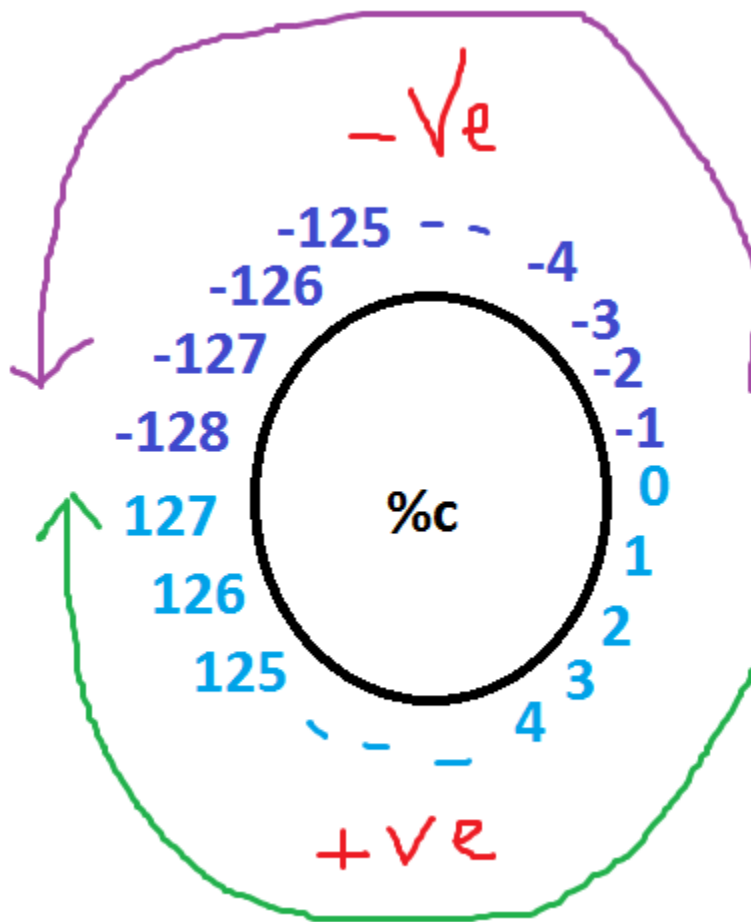


A screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" and standard Windows window controls. The main area is black with white text showing three lines of code: `c=-125`, `c=4`, and `c=125`. In the bottom right corner of the window, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 12:02 PM on 19-Feb-25.

C & C++ are using ASCII character set, 256 characters

256 characters $\left\{ \begin{array}{l} \text{signed char} \Rightarrow -128 \text{ to } +127 \\ \text{unsigned char} \Rightarrow 0-255 \end{array} \right.$

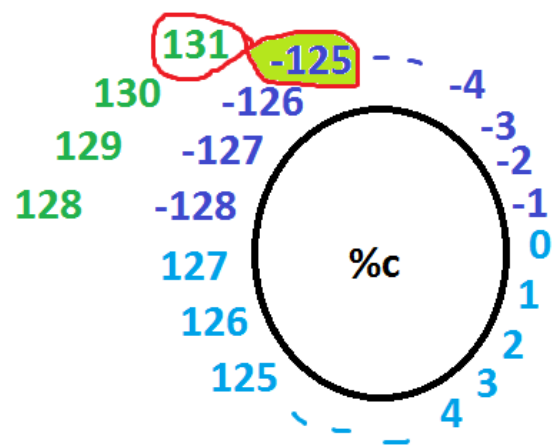
Char cycle:



a=65
b=66
c=131

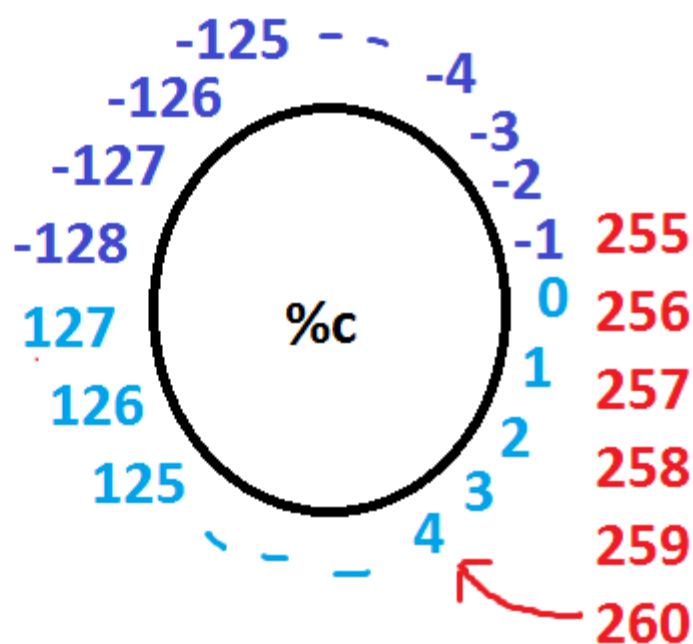
$$\begin{array}{r} 131 \\ -128 \\ \hline 3 \end{array} + 3 = -125$$

$$\begin{array}{r} 256 \\ -131 \\ \hline -125 \end{array}$$



c = 260;

260-256=4



256

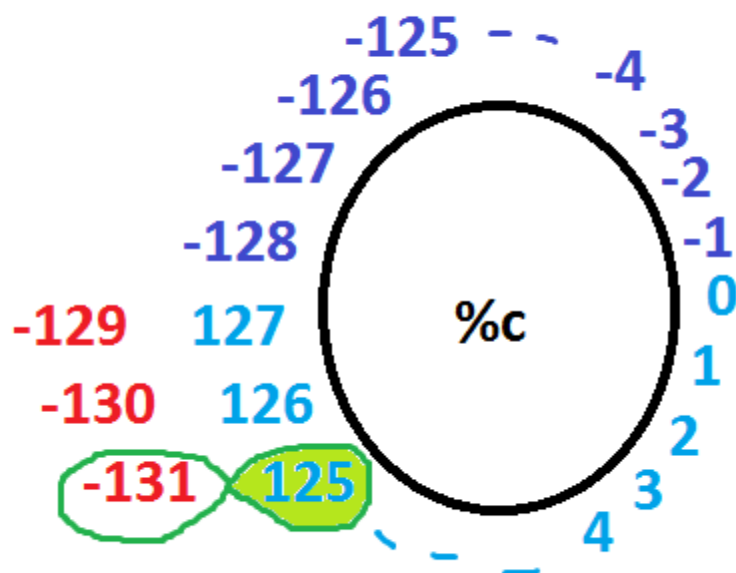
-131

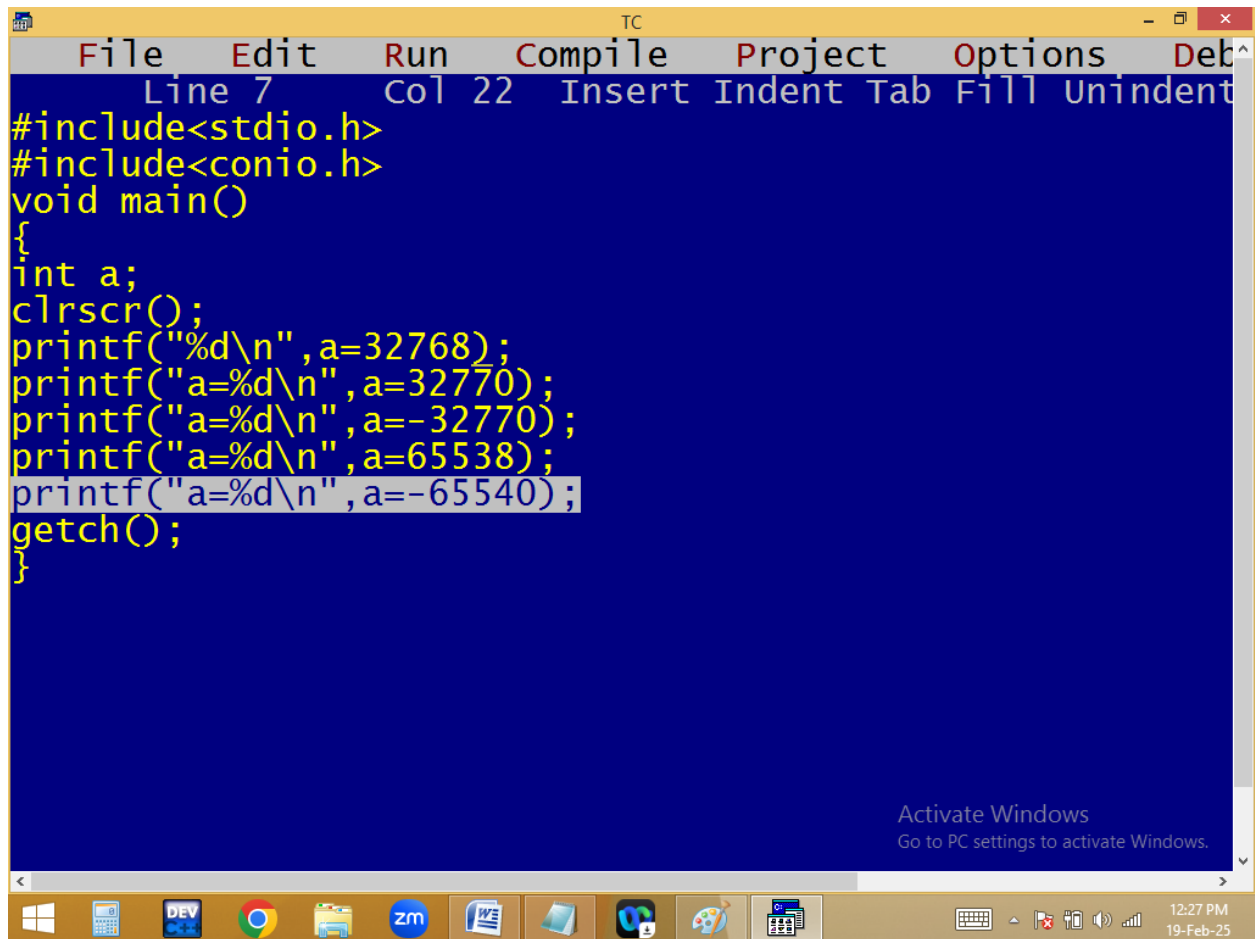
125

-131

128-3=125

-3





```
TC
File Edit Run Compile Project Options Deb
Line 7 Col 22 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
printf("%d\n",a=32768);
printf("a=%d\n",a=32770);
printf("a=%d\n",a=-32770);
printf("a=%d\n",a=65538);
printf("a=%d\n",a=-65540);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

12:27 PM
19-Feb-25

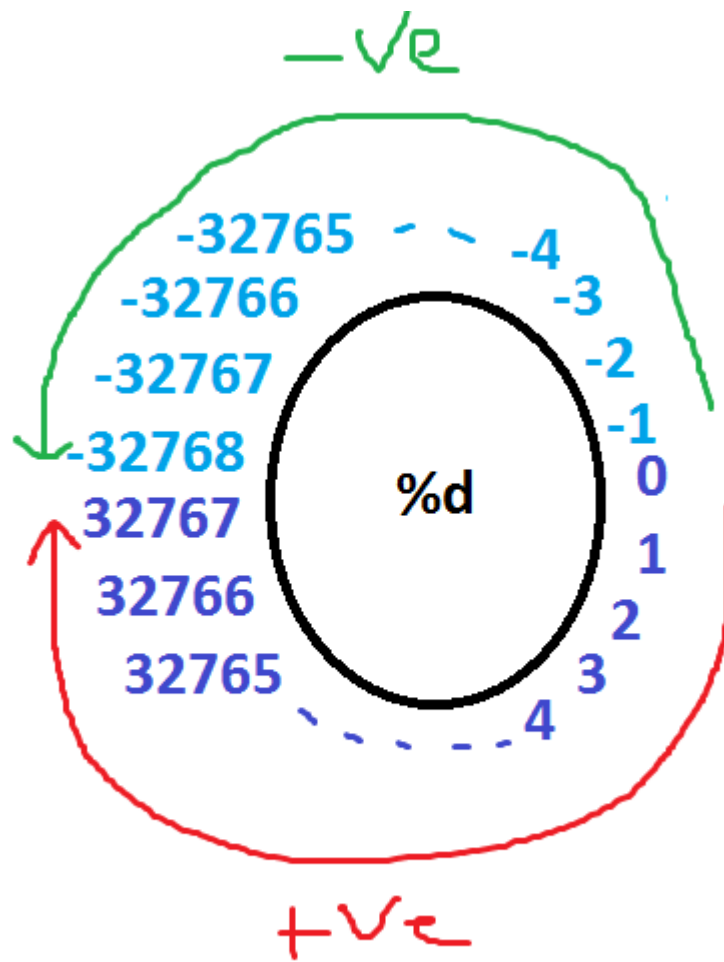
The screenshot shows a Turbo C++ (TC) compiler window with a black background and white text. The text displays the following code:
-32768
a=-32766
a=32766
a=2
a=-4
A horizontal cursor line is positioned below the last line of code. The window's title bar is yellow and contains the text 'TC'. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 12:27 PM on 19-Feb-25. An 'Activate Windows' watermark is present in the bottom right corner of the window.

C & C++ are using 16 bit compilers i.e. $2^{16} = 65536$

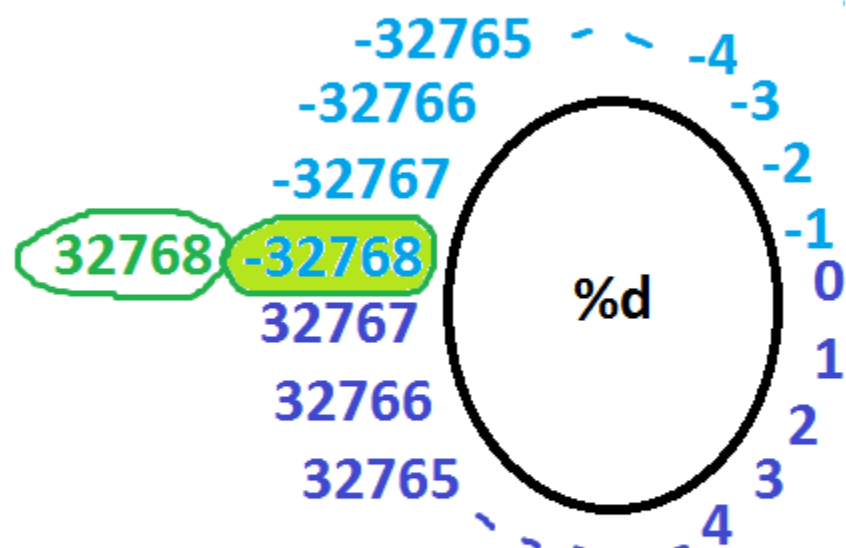
65536 {
 signed int ==> -32768 to +32767
 unsigned int ==> 0 to 65535

java/ .net or online compiler ==> $2^{32} ==> 4294967296$ {
 signed
 -2147483648 to +2147483647
 unsigned
 0 to 4294967295

Signed int cycle:

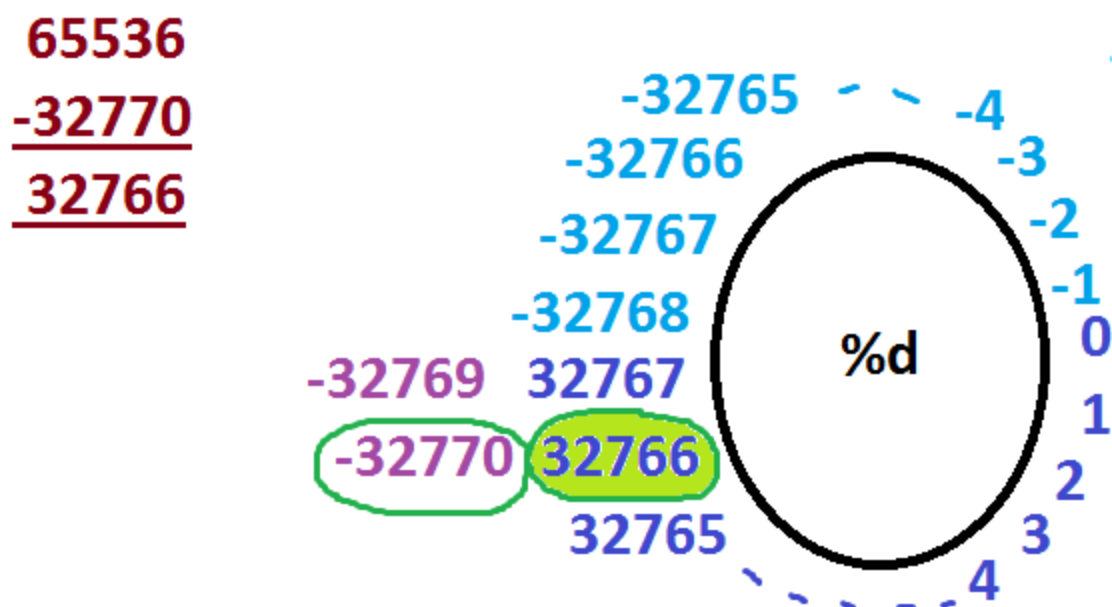
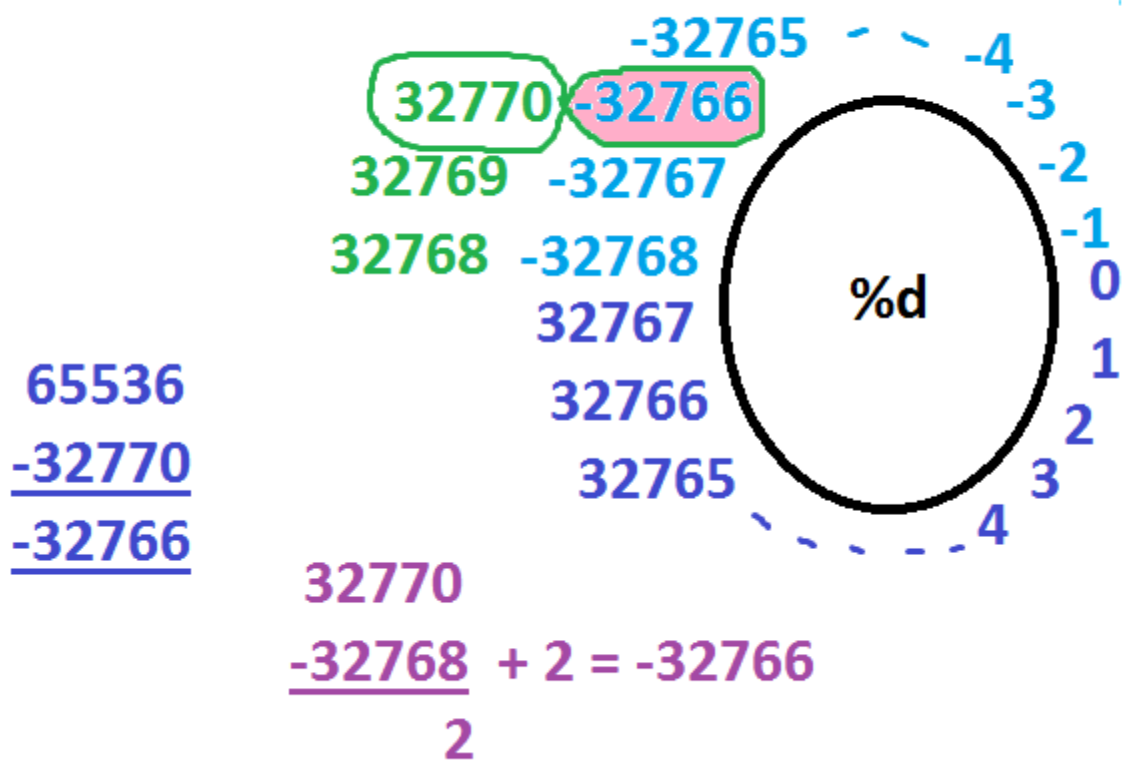


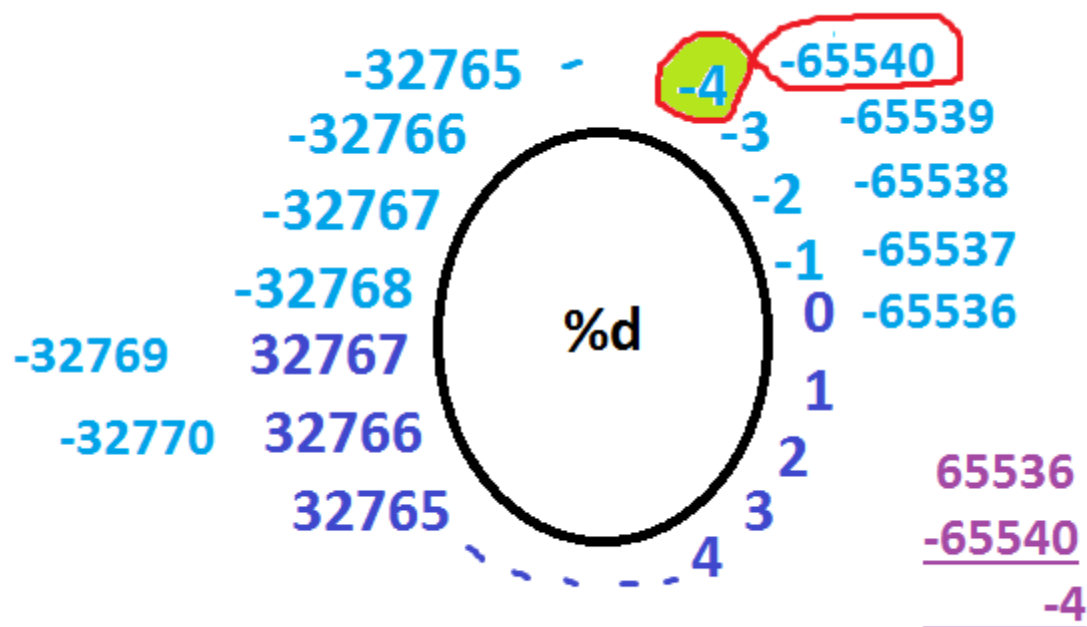
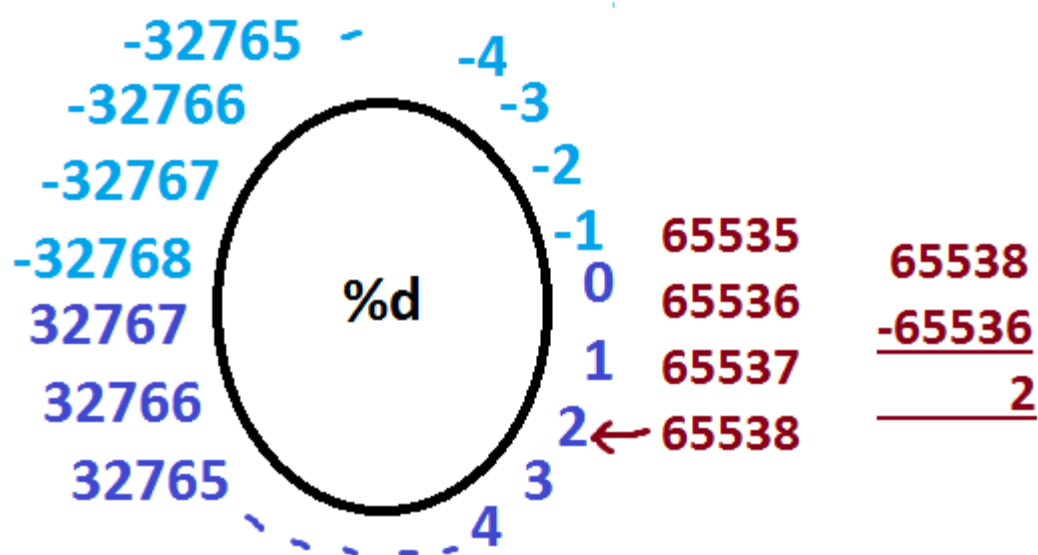
a=32768



65536
-32768
-32768

32768
-32768 + 0 = -32768
0





$$\begin{array}{r}
 532 \quad 1 \quad 4 \quad 0 \quad 1 \quad 3 \quad 9 \\
 533 \quad 1 \quad 3 \quad 1 \\
 534 \quad 2 \quad 3 \quad 1 \\
 \hline
 65535 \quad 1 \quad 3 \quad 7 \\
 \hline
 \square \quad - \quad 65536
 \end{array}$$

$$\begin{array}{r} 32 \text{ } 44 \text{ } 1 \text{ } 39 \\ 538 \\ 534 \text{ } -2 \text{ } 31 \\ \hline 65535 \text{ } -1 \text{ } 37 \\ \hline \square \text{ } -65536 \end{array}$$